2026

探路者：虚实融合数智具身平台

VR 沉浸式遥操作 · VLA 任务规划与预检 · FAST-LIO2 建图导航

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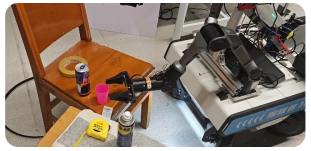


01 项目简介

PROJECT OVERVIEW

面向远程人机协作与移动操作，将 VR 沉浸式遥操作、本地语言任务规划和雷达惯性建图集成于移动双臂机器人。

典型应用与场景反馈







千问 Qwen

YOLO

遥操作抓取 · 多路视频 · 三维点云 · 本地大模型

关键技术

01 VR 沉浸式交互
双手离合增量映射、One Euro 滤波与分层解析 IK

02 FAST-LIO2 点云建图
雷达惯性融合、几何定位与彩色 / 轻量场景生成

03 大模型语言任务
本地千问、YOLO 目标定位、轨迹预检与人工授权

02 创新设计

系统架构与共享执行底座

Quest 3 | 相机与雷达 | 工作台

↓

AMD Ryzen AI Max+ 395

VR 映射与 IK | 千问 + YOLO | FAST-LIO2

导航地图 · 语义证据 · 全量 / 轻量场景

↓

KR260 · PS / PL

硬件队列 · CAN · 双以太网 · 反馈快照 · ILA

↓

双六轴机械臂 | 四转四驱底盘

AI PC 推理决策与任务调度

本地推理：千问与 YOLO 完成语言解析、目标定位及动作候选生成。
规划预检：轨迹生成、关节限位与碰撞检查，人工授权后下发。
感知调度：VR 映射、FAST-LIO2 建图导航与彩色点云异步协同。

KR260 控制安全中枢

校验执行： PS/Linux 完成目标校验、限速插值、会话管理与软件停止。
硬件下发： PL 组织硬件命令队列、反馈快照及双路 CAN / 以太网接口。
通路保护：通路切换、端口恢复与 ILA 在线观测支撑安全执行。

03 结果与展望

VR 控制采样至编码器反馈

148.08 ms — P50 · N=54

P95 175.46 ms · P99 191.81 ms

语言抓取 · 限定工作区

92% — 46 / 50

50 次完成 46 次

建图定位与导航执行

5.14 / 5.90 cm — 离线 RMSE

2 个留出扫描；导航 19 条：10 条到达，3 条人工停止

KR260 执行底座与 FPGA 实现

20 ms — PS 反馈周期

双路恢复后 3 s：9,120 / 9,130 条 CAN 帧
LUT 41.28% · 寄存器 31.79% · BRAM 89.24% · DSP 0%
WNS +0.082 ns · DRC 错误 0

彩色场景与计算

全量 3994 万点，轻量 57.96 万点（5 cm）。
8 图像着色任务：ROCm 双进程相对 CPU 四进程加速 1.864×。

TCP 通信

双向各 20.00 Mbit/s × 60 s，SHA-256 校验通过。

拓展：自主探索 · 长时 VR · 双臂协作 · 指令泛化

2026

PathFinder: Embodied Intelligence Platform

Immersive VR Teleoperation · VLA Planning & Prechecks · FAST-LIO2 Mapping & Navigation

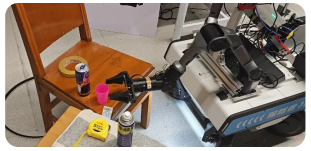
Members: Ziang Li · Chengchen Ding · Kunpeng Zhou
Advisor: Liqing Zhou
Institution: Wuhan University



01 Project Overview

For remote human–robot collaboration and mobile manipulation, PathFinder integrates immersive VR teleoperation, on-device language task planning, and LiDAR-inertial mapping on a mobile dual-arm robot.

Applications & Scene Feedback







千问 Qwen

YOLO

Teleoperated Grasping · Multi-view Video · 3D Point Cloud · On-device AI

Key Technologies

01 Immersive VR Interaction
Bimanual clutch-based incremental mapping, One Euro filtering & hierarchical analytic IK

02 FAST-LIO2 Point-cloud Mapping
LiDAR-inertial fusion, geometric localization, colorized & lightweight scenes

03 Language-driven AI Tasks
On-device Qwen, YOLO localization, trajectory prechecks & human authorization

02 System & Innovation

System Architecture & Shared Execution Platform

Quest 3 | Camera & LiDAR | Workstation

↓

AMD Ryzen AI Max+ 395

VR Mapping & IK | Qwen + YOLO | FAST-LIO2

Navigation Maps · Semantic Evidence · Full / Lightweight Scenes

↓

KR260 · PS / PL

Hardware Queues · CAN · Dual Ethernet · Feedback Snapshots · ILA

↓

Dual 6-axis Arms | 4-wheel Steer & Drive Base

AI PC Inference, Decision-making & Task Orchestration

On-device inference: Qwen and YOLO parse commands, localize targets and generate action candidates.
Planning prechecks: Trajectory generation, joint-limit and collision checks before human-authorized dispatch.
Perception scheduling: Asynchronous VR mapping, FAST-LIO2 navigation and point-cloud colorization.

KR260 Safety-critical Control Hub

Validated execution: PS/Linux validates targets, interpolates with speed limits, and manages sessions and software stop.
Hardware dispatch: PL implements command queues, feedback snapshots, dual CAN and Ethernet interfaces.
Path protection: Path switching, port recovery and online ILA observation support safe execution.

03 Results & Outlook

VR Command Sample to Encoder Feedback

148.08 ms — P50 · N=54

P95 175.46 ms · P99 191.81 ms

Language-guided Grasping · Bounded Workspace

92% — 46 / 50

46 successful grasps out of 50 trials

Mapping, Localization & Navigation

5.14 / 5.90 cm — Offline RMSE

2 held-out scans; 19 navigation runs: 10 arrivals, 3 manual stops

KR260 Execution Platform & FPGA Implementation

20 ms — PS Feedback Cycle

3 s after dual-path recovery: 9,120 / 9,130 CAN frames
LUT 41.28% · FF 31.79% · BRAM 89.24% · DSP 0%
WNS +0.082 ns · 0 DRC Errors

Colorized Scenes & Computing

Full scene: 39.94M points; lightweight: 579.6k points (5 cm).
8 colorization tasks: dual-process ROCm is 1.864× faster than quad-process CPU.

TCP Communication

20.00 Mbit/s each way for 60 s; SHA-256 verification passed.

Next: Autonomous Exploration · Extended VR · Dual-arm Collaboration · Instruction generalization